DANG-Med: Dual-Stream Attention-Augmented Network with Prototype-Anchored Spatial Support and Gated Directional Strip Alignment for Multi-Center Polyp Segmentation

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Abstract

Deep convolutional and vision transformer networks have achieved strong in-domain accuracy on curated colonoscopy benchmarks such as Kvasir-SEG ($\sim$ 93% mDice), yet their segmentation reliability degrades under real-world clinical stress conditions: (i) severe label scarcity when adapting to new clinical centers (5%–10% labeled frames), (ii) natural tiny ($< 1.5\%$ mucosal area) and flat low-contrast camouflaged polyps across unseen imaging systems (e.g., ETIS-LaribPolypDB), and (iii) adverse endoscopic acquisition shifts including Narrow-Band Imaging (NBI) spectral emphasis, distal-lumen low-light falloff, and wet mucosal specular glare with motion defocus. To address these clinical failure modes on a single GPU, we propose **DANG-Med** (*Dual-Stream Attention-Augmented Network with Prototype-Anchored Spatial Support and Gated Directional Strip Alignment*), a hybrid architecture coupling a local multi-scale ConvNeXt-Tiny hierarchy with a frozen DINOv2-Base foundation backbone (86.58M frozen parameters) adapted via Frequency-Routed Mixture-of-Experts Convolutional LoRA (**Freq-MoE Conv-LoRA**). DANG-Med introduces four architectural pillars: (1) a **4-Scale Learned DPT Multi-Layer Reassembler** with a **Dual-Prototype and Image-Adaptive Self-Support Margin Head** that separates polyp versus mucosa token manifolds under 40-image few-shot supervision; (2) a **Prototype-Anchored Spatial Support Envelope (PASSE)** that suppresses isolated far-field false-positive background specks while reinforcing detected lesion cores; (3) a **Multi-Scale Homomorphic Retinex Chromatic Stem** paired with a **DINOv2-Conditioned Low-Frequency Style Calibrator** (with feature-space MixStyle); and (4) an **Uncertainty-Guided Reverse-Attention Directional Strip Decoder** (**FastAxialStripSSM**) modulated by normalized Bernoulli boundary variance $4p(1-p)$. We evaluate two protocols across deduplicated in-domain, multi-center out-of-distribution (PolypGen 6-center Dev $n = 258$ and Extended $n = 974$, CVC-ClinicDB, CVC-ColonDB, CVC-300, ETIS-LaribPolypDB), and synthetic adverse stress cohorts: (A) our primary **Round 9/10 zero-shot stem protocol** (5% few-shot, $n = 40$, 5 seeds 42–46), where all models share the same primary segmentation objective and no adverse synthetic corruptions are used during training (`aug_prob=0.0`); and (B) our **Round 8 protocol** (10% few-shot, $n = 80$, Seed 42), where `R8e` activates in-family online adverse synthesis (`aug_prob=0.55`). Across all 5 paired seeds (Seeds 42–46) under the 5% few-shot protocol, **DANG-Med v4.1 Flagship (`R9e+R10d/e`)** improves over the single-stream ConvNeXt-Tiny+FPN baseline (`R9a+R10a`) from $44.74 \pm 3.79\%$ to $51.05 \pm 1.71\%$ **mDice (+6.31 pp paired mean,** $p = 0.014$**; Seed 42: +10.79 pp)** and from $16.22 \pm 3.87\%$ to $30.65 \pm 1.16\%$ **Boundary F1 (+14.44 pp,** $p = 0.002$**; Seed 42: +22.68 pp)** on few-shot tiny/camouflaged lesions ($S_1 \times S_2$, $n = 310$); from $45.59 \pm 4.02\%$ to $55.82 \pm 1.17\%$ **mDice (+10.23 pp,** $p = 0.005$**)** on zero-shot adverse endoscopy ($S_1 \times S_3$, $n = 258$); and from $66.59 \pm 3.84\%$ to $71.86 \pm 0.34\%$ **mDice (+5.27 pp,** $p = 0.038$**; Seed 42: +11.86 pp)** and $28.45 \pm 4.82\%$ to $42.25 \pm 1.58\%$ **BF1 (+13.80 pp)** on external PraNet benchmarks ($S_1 \rightarrow$ PraNet-Ext, $n = 696$). Across all 5 paired seeds on full-data training ($n = 800$), zero-shot **DANG-Med v4.1** reaches $78.88 \pm 1.50\%$ **mDice on ETIS-LaribPolypDB** (vs. $72.55 \pm 2.68\%$ baseline, +6.33 pp, $p = 0.005$; Seed 42: 80.78%) and $81.82 \pm 0.64\%$ **on PraNet-Ext** (vs. $78.66 \pm 1.55\%$, $p = 0.003$), while Round 8 **DANG-Med v4 (`R8e`)** reaches 81.53% **mDice on ETIS-LaribPolypDB** and 73.87% **mDice on corruption-matched Adverse Endoscopy (S_3, $n = 272$)**.

1 Introduction

Colorectal cancer (CRC) remains one of the leading causes of cancer-related mortality worldwide, and early endoscopic detection and resection of adenomatous and sessile serrated polyps during optical colonoscopy significantly reduces CRC incidence (Jha et al., 2020; Ali et al., 2023). Automated pixel-level polyp segmentation provides computer-aided delineation (CADe/CADx) of lesion mar-

gins during withdrawal. However, a substantial gap persists between in-domain benchmark saturation and multi-center clinical deployment. On standard in-domain white-light colonoscopy splits such as Kvasir-SEG (Jha et al., 2020), modern hierarchical convolutional networks (e.g., ConvNeXt-Tiny (Liu et al., 2022) with Feature Pyramid Networks (Lin et al., 2017)) already achieve 92.5%–93.1% mean Dice (mDice). Under such saturated in-domain conditions, architectural modifications yield modest differences ($<$ 0.5–1.0 percentage points), obscuring whether a proposed model genuinely improves out-of-distribution (OOD) robustness.

In clinical practice, endoscopic segmentation systems degrade across three compounding stress axes:

1. **Label-Scarce Multi-Center Adaptation (S_1):** Annotating pixel-accurate polyp contours requires expert gastroenterologists. When training under a strict 5%–10% annotation budget (40–80 labeled images), standard CNNs overfit to source-hospital background textures and suffer severe boundary degradation on external centers (Boundary F1 $<$ 20%, HD95 $>$ 70 px).

2. **Natural Tiny and Low-Contrast Camouflaged Lesions (S_2):** Flat Paris 0-IIa/IIb adenomas and diminutive polyps occupying $<$ 1.5% of the endoscopic field of view (FOV) or exhibiting low RGB contrast ($<$ 0.085) against surrounding vascular mucosa are frequently missed or fragmented by stride-16/32 feature pooling, particularly on external cohorts such as ETIS-LaribPolypDB (Silva et al., 2014) and CVC-ColonDB (Tajbakhsh et al., 2016).

3. **Adverse Endoscopic Acquisition Shifts (S_3):** Endoscopic procedures frequently switch between White-Light Imaging (WLI) and Narrow-Band Imaging (NBI)-like spectral shifts, encounter distal-lumen illumination falloff ($\gamma > 2.2$ with sensor noise), and suffer from wet mucosal specular highlights coupled with motion/defocus blur.

While Vision Foundation Models (VFMs) such as DINOv2 (Oquab et al., 2024) and SAM (Kirillov et al., 2023) offer strong semantic invariance across domains, their coarse patch grids (14×14 stride) lack fine subpixel boundary localization. Conversely, naive feature concatenation between a CNN encoder and a ViT backbone can degrade high-resolution CNN edge features when ViT patch tokens are bilinearly upsampled across lesion boundaries, and sequential 2D state-space scans (SS2D (Liu et al., 2024)) incur substantial Python/CUDA kernel launch overhead on multi-scale feature pyramids.

In this work, we present **DANG-Med**, illustrated in Figure 1. Our main contributions are:

- **Auditable Multi-Center Deduplication Hygiene and Clinical Stress Evaluation:** We implement an automated perceptual-hash and MD5 deduplication pipeline (`HYGIENE_MANIFEST_JSON`) across Kvasir-SEG, 6-center PolypGen (Ali et al., 2023) (Dev $n = 258$ and untouched Extended Holdout $n = 974$ in Round 9/10; $n = 272$ and $n = 967$ in Round 8), all 4 official PraNet external benchmarks (Fan et al., 2020) ($n = 696$ after removing 2 exact/FOV duplicates against Kvasir: CVC-300 $n = 60$, CVC-ClinicDB $n = 61$, CVC-ColonDB $n = 379$, ETIS-LaribPolypDB $n = 196$), and derived stress slices (S_1 5%/10% few-shot, S_2 pooled tiny/camouflaged lesions $n = 310/315$, and S_3 synthetic adverse acquisitions $n = 258/272$).

- **4-Scale DPT Reassembly, Dual-Prototype Self-Support, and PASSE Envelope:** We project multi-depth DINOv2 tokens into an L_2-normalized prototype hypersphere combining orthogonally initialized K-prototype cosine margins, training-gated ground-truth EMA support anchors, and test-time image-adaptive self-support centroids. Crucially, we introduce the **Prototype-Anchored Spatial Support Envelope (PASSE)**, which forms a dilated semantic support envelope to suppress far-field false-positive background specks on detected lesions while reinforcing low-contrast polyp cores.

- **Retinex Chromatic Stem and DINO-Conditioned Style Calibration:** We design a differentiable homomorphic Retinex stem (`RetinexChromaticStem`) paired with DINOv2-conditioned low-frequency style calibration and feature-space MixStyle at strides 4 and 8. We evaluate this module both under a strict zero-shot stem regime in Round 9/10 (`aug_prob=0.0`, where the stem is performance-neutral relative to `ProtoDPT+PASSE` alone) and under corruption-matched online shift synthesis in Round 8 (`aug_prob=0.55`).

- **Directional Strip Alignment and Uncertainty-Guided Reverse-Attention Decoder:** We formulate **FastAxialStripSSM** using fused cuDNN horizontal ($1\times11, 7\times1$), vertical ($11\times1, 1\times7$), and dilated ($3\times3, d=3$) depthwise strip operators modulated by normalized Bernoulli boundary variance $u = 4p(1-p)$ and outer morphological ring halos.

2 Related Work

2.1 Polyp Segmentation and Boundary-Aware Decoders

Deep polyp segmentation architectures have progressed from U-Net (Ronneberger et al., 2015) and UNet++ (Zhou et al., 2018) to reverse-attention and boundary-aware networks. PraNet (Fan et al., 2020) introduced parallel partial decoders and iterative reverse attention to mine boundary cues from Res2Net backbones. CaraNet (Lou et al., 2022) extended reverse attention with axial context and channel-wise feature pyramid modules for small medical objects. Subsequent Transformer-based architectures such

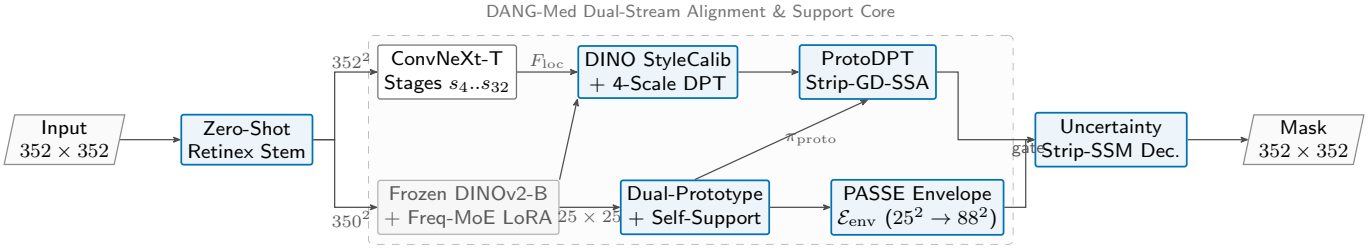


Figure 1: **Overview of the proposed DANG-Med architecture.** An input endoscopic frame (352 × 352) passes through the Multi-Scale Homomorphic Retinex Chromatic Stem (`RetinexChromaticStem`) to suppress specular highlights and standardize illumination/chrominance against source white-light EMA statistics. The local stream (ConvNeXt-Tiny stages $s_4..s_{32}$) captures sharp mucosal edges while the semantic stream (frozen DINOv2-Base with Freq-MoE Conv-LoRA on blocks 8..11) extracts multi-depth 25 × 25 patch tokens from layers $\{2, 5, 8, 11\}$. Our highlighted contributions (blue) include DINO-Conditioned Low-Frequency Style Calibration, 4-Scale Learned DPT Reassembly, the Dual-Prototype & Self-Support Margin Head, the Prototype-Anchored Spatial Support Envelope (PASSE), and the Uncertainty-Guided Reverse-Attention Directional Strip Decoder.

as Polyp-PVT (Dong et al., 2021) and TransFuse (Zhang et al., 2021) paired or replaced CNN encoders with Pyramid Vision Transformers to improve multi-center generalization on CVC-ColonDB and ETIS-LaribPolypDB. Unlike dual-branch models that fuse features only at coarse 1/16 and 1/32 scales with unconstrained bilinear upsampling, DANG-Med performs 4-scale learned DPT depth mixing with content-aware edge-snapped upsampling and explicit spatial support gating at stride 4 (88 × 88).

2.2 Vision Foundation Models and Parameter-Efficient Adaptation

Self-supervised Vision Transformers pretrained via discriminative self-distillation, notably DINOv2 (Oquab et al., 2024), exhibit strong part-level semantic consistency across domains, while promptable foundation models such as SAM (Kirillov et al., 2023) and its medical adaptations require either bounding-box prompts or heavy decoder fine-tuning. Adapting VFMs to autonomous dense medical segmentation while preventing overfitting on small clinical datasets ($N \leq 800$) benefits from parameter-efficient fine-tuning (PEFT) such as LoRA (Hu et al., 2022) and Dense Prediction Transformers (DPT) (Ranftl et al., 2021). We freeze 100% of the DINOv2-Base backbone (86.58M frozen parameters logged across all dual-stream runs) and inject lightweight Frequency-Routed Mixture-of-Experts Convolutional LoRA (**Freq-MoE Conv-LoRA**) adapters into blocks 8–11, decomposing spatial tokens into low-frequency context and high-frequency edge energy routed across dilation rates $d \in \{1, 2, 3\}$.

2.3 State-Space and Directional Strip Operators in Medical Vision

Structured State-Space Sequence Models (S4, Mamba (Gu and Dao, 2023)) and 2D visual extensions (VMamba (Liu et al., 2024), U-Mamba (Ma et al., 2024)) model long-range spatial dependencies. However, executing recursive 4-directional 1D scans in Python across multi-scale 2D feature pyramids incurs substantial kernel launch overhead. Motivated by the connection between truncated state-space convolutional kernels and directional strip convolutions, our **FastAxialStripSSM** captures horizontal, vertical, and dilated contextual dependencies using fused depthwise strip convolutions gated by cross-stream conditioning, achieving 17.4–20.0 FPS (measured at batch size 1, 352 × 352, FP16 on NVIDIA T4).

3 Methodology: The DANG-Med Architecture

Given an input endoscopic RGB image $I \in \mathbb{R}^{B\times3\times H\times W}$ ($H = W = 352$), DANG-Med (`src/dang/model.py`) predicts a dense foreground polyp probability map $\hat{Y} = \sigma(Z) \in [0, 1]^{B\times1\times H\times W}$. Figure 1 illustrates the complete pipeline.

3.1 Retinex Chromatic Stem and DINO-Conditioned Style Calibrator

Endoscopic frames from external centers or adverse modes exhibit non-uniform illumination and spectral shifts. Let $X_{\text{rgb}} \in [0, 1]^{B\times3\times H\times W}$ denote the unnormalized RGB image entering `RetinexChromaticStem` (instantiated in code whenever `use_style_calib=True`).

1) Specular Highlight Inpainting: Mucosal glare saturates all three RGB channels. We detect specular pixels via a smooth minimum-channel gate $w_{\text{glare}} = \sigma(18.0 \cdot (\min_c X_{\text{rgb}} - 0.82))$ and blend them with a 15 × 15 local average filter $\bar{X}_{15}$:

$$X_{\text{deg}} = (1 - w_{\text{glare}})X_{\text{rgb}} + w_{\text{glare}}\bar{X}_{15}. \quad (1)$$

2) Multi-Scale Homomorphic Retinex Division: Approximating image intensity as the product of illumi-

nation $L(x)$ and mucosal reflectance $R(x)$, we estimate illumination envelopes $L_{31} = \mathrm{AvgPool}_{31\times31}(X_{\deg}) + \epsilon$ and $L_{15} = \mathrm{AvgPool}_{15\times15}(X_{\deg}) + \epsilon$ ($\epsilon = 0.03$) and recover log-reflectance:

$$\tilde{R} = \log(X_{\deg} + \epsilon) - \big(0.65 \log L_{31} + 0.35 \log L_{15}\big). \quad (2)$$

3) Source EMA Standardization and Anomaly Gate: During training on source Kvasir-SEG images, the stem updates exponential moving average reference buffers $(\mu_{\mathrm{ref}}, \sigma_{\mathrm{ref}}) \in \mathbb{R}^{1\times3\times1\times1}$ (in training mode only; frozen during `model.eval()`). In our Round 9/10 protocol (`R9d`, `R9e`, `R10d`, `R10e`), `aug_prob` inside the stem is hard-set to 0.0 so no synthetic adverse corruptions are generated during training (source-trained with mild photometric jitter $\gamma \in [0.8, 1.25]$, evaluated zero-shot on unseen S_3 synthetic corruptions where $\gamma = 2.55$). In Round 8 (`R8e`, commit `9539e96`), `aug_prob=0.55` synthesizes NBI, low-light ($\gamma = 2.35$), glare+blur, and combined corruptions with 0.55 contrast compression during training; thus `R8e`'s S_3 score reflects in-family corruption-matched training rather than zero-shot OOD generalization. Combining instance-standardized reflectance $\hat{R}$ and global chromaticity $\hat{X}_{\deg}$, the restored image is $X_{\mathrm{rest}} = \mathrm{clip}((0.55\hat{X}_{\deg} + 0.45\hat{R})\sigma_{\mathrm{ref}} + \mu_{\mathrm{ref}}, 0, 1)$. A photometric anomaly score A_{phot} measures the normalized deviation of $(\mu(X_{\deg}), \sigma(X_{\deg}))$ from $(\mu_{\mathrm{ref}}, \sigma_{\mathrm{ref}})$ plus glare fraction, gating a learned 3×3 residual refinement branch (with minimum blend floor 0.04).

At high-resolution ConvNeXt stages s_4 (88×88) and s_8 (44×44), **DINOFreqStyleCalibrator** first applies feature-space MixStyle (Zhou et al., 2018) with probability $p = 0.5$ (Beta(0.1, 0.1) instance-stat mixing across the batch during training only) and then decomposes local CNN features F_{loc} into a 9×9 low-frequency component F_{low} and high-frequency detail $F_{\mathrm{high}} = F_{\mathrm{loc}} - F_{\mathrm{low}}$, modulating both via affine parameters $(\gamma_{\mathrm{low}}, \beta_{\mathrm{low}}, \gamma_{\mathrm{high}})$ predicted from the global DINOv2 context vector $g_{\mathrm{dino}} \in \mathbb{R}^{B\times768}$.

3.2 4-Scale DPT Reassembly and Dual-Prototype Self-Support Margin Head

We extract spatial token grids $\{T_\ell\}_{\ell\in\{2,5,8,11\}} \subset \mathbb{R}^{B\times768\times25\times25}$ from DINOv2-Base (with Freq-MoE Conv-LoRA of rank $r = 4$ on blocks 8–11) and project each to $D = 192$ channels: $S_\ell = \mathrm{ConvBNAct}_{1\times1}(T_\ell)$.

4-Scale Learned DPT Reassembler: We learn a 4×4 depth-mixing matrix $W = \mathrm{Softmax}(\Theta) \in \mathbb{R}^{4\times4}$ across scales $s \in \{0, 1, 2, 3\}$:

$$\tilde{S}_s = \sum_{\ell=0}^{3} W_{s,\ell} S_\ell, \quad \hat{S}_s = \phi_s\big(\mathrm{Up}_s(\tilde{S}_s, F_{\mathrm{loc}}^{(s)})\big), \quad (3)$$

where scales s_8 and s_{16} use content-aware **Edge-Snapped Semantic Upsampling** (predicting 3×3 local reassembly kernels from CNN features $F_{\mathrm{loc}}^{(s)}$, initialized to bilinear interpolation).

Dual-Prototype and Self-Support Margin Head: We concatenate $[S_1, S_2, S_3]$, project to $D_p = 128$ dimensions, and L_2-normalize across channels to obtain unit token embeddings $\hat{Z} \in \mathbb{R}^{B\times D_p\times25\times25}$. We initialize $K = 8$ learnable foreground prototypes $\{p_k^+\}_{k=1}^K$ and $K = 8$ background prototypes $\{p_k^-\}_{k=1}^K$ orthogonally, alongside training-gated ground-truth EMA support anchors (a^+, a^-) updated only during training from downsampled mask occupancy ($M_{25} \geq 0.35$ for foreground, $M_{25} \leq 0.10$ for background). The prior cosine margin logit at each 25×25 patch (u, v) is:

$$\pi_{\mathrm{prior}}(u,v) = \tau\Big(\frac{1}{\beta}\log\sum_{k=1}^{K} e^{\beta\langle\hat{Z}_{u,v},\hat{p}_k^+\rangle} - \frac{1}{\beta}\log\sum_{k=1}^{K} e^{\beta\langle\hat{Z}_{u,v},\hat{p}_k^-\rangle}\Big) + w_{\mathrm{ema}} \cdot 10\big(\langle\hat{Z}_{u,v}, a^+\rangle - \langle\hat{Z}_{u,v}, a^-\rangle\big) + b. \quad (4)$$

During inference (without ground-truth masks), we compute image-specific self-support centroids $c^+ = \mathrm{Norm}(\sum_{u,v} p_{u,v}^2 \hat{Z}_{u,v})$ and $c^- = \mathrm{Norm}(\sum_{u,v}(1-p_{u,v})^2 \hat{Z}_{u,v})$ where $p_{u,v} = \sigma(\pi_{\mathrm{prior}}(u, v))$, yielding the 25×25 prototype margin map:

$$\pi_{\mathrm{proto}}(u,v) = \pi_{\mathrm{prior}}(u,v) + 8w_{\mathrm{ss}}\big(\langle\hat{Z}_{u,v}, c^+\rangle - \langle\hat{Z}_{u,v}, c^-\rangle\big). \quad (5)$$

3.3 Prototype-Anchored Spatial Support Envelope (PASSE)

While prototype priors strengthen semantic localization on detected lesions, uncalibrated high-frequency CNN texture noise far from the true polyp can cross the 0.5 threshold, inflating false positives and HD95. To suppress far-field false positives without eroding lesion interiors, **PASSE** constructs a 25×25 semantic probability map $P_{\mathrm{sem}} = \sigma(0.5M_4^{\uparrow25} + 0.5\pi_{\mathrm{proto}})$ from the coarsest decoder stage logit M_4 and prototype logit π_{proto} (falling back to $P_{\mathrm{sem}} = \sigma(M_4^{\uparrow25})$ when the prototype head is disabled, as in variant `R9d`), and computes a dilated smooth support envelope:

$$\mathcal{E}_{\mathrm{env}} = \mathrm{AvgPool}_{3\times3}\big(\mathrm{MaxPool}_{5\times5}(P_{\mathrm{sem}})\big) \in [0,1]^{B\times1\times25\times25}. \quad (6)$$

Upsampling $\mathcal{E}_{\mathrm{env}}$ and P_{sem} to stride 4 (88×88), PASSE modulates the stride-4 decoder logit Z_{88} via two spatial terms, after which the final output logit blends the refined decoder prediction with the upsampled prototype margin when the prototype head is active:

$$Z_{88}^{\mathrm{PASSE}} = Z_{88} - 2.2\,\frac{\mathrm{ReLU}(0.22 - \mathcal{E}_{\mathrm{env}}^{\uparrow88})}{0.22} + 1.2\,\mathrm{ReLU}(P_{\mathrm{sem}}^{\uparrow88} - 0.38),$$
$$Z = (1 - 0.2\,\mathbb{I}[\mathrm{proto}])\big(Z_{88}^{\mathrm{PASSE}}\big)^{\uparrow352} + 0.2\,\mathbb{I}[\mathrm{proto}]\,\pi_{\mathrm{proto}}^{\uparrow352}. \quad (7)$$

3.4 Uncertainty-Guided Directional Strip Decoder

At each pyramid scale $s \in \{s_8, s_{16}, s_{32}\}$, **GDSSAFusion** aligns local CNN features $F_{\mathrm{loc}}^{(s)}$ and semantic features $\hat{S}_s$ us-

ing **FastAxialStripSSM**, a gated finite-kernel directional strip operator:

$$\begin{aligned}\mathrm{StripSSM}(U, C) = U &+ g_h \odot \mathcal{K}_{7\times1}(\mathcal{K}_{1\times11}(U)) \\ &+ g_v \odot \mathcal{K}_{1\times7}(\mathcal{K}_{11\times1}(U)) + g_d \odot \mathcal{K}_{3\times3,d=3}(U),\end{aligned} \tag{8}$$

where $(g_h, g_v, g_d) = \sigma(\mathrm{Conv}_{1\times1}([U, C]))$ are input-conditioned directional gates and $\mathcal{K}$ denotes depthwise strip convolutions. In the top-down decoder ($P_4 \to P_3 \to P_2 \to P_1$), each **UncertaintyStripDecoderStage** computes from the coarser stage's logit $M_{s+1}^{\uparrow}$ three spatial priors: (i) foreground probability $A_{\mathrm{fg}} = \sigma(M_{s+1}^{\uparrow})$, (ii) normalized Bernoulli boundary variance $U_s = 4A_{\mathrm{fg}}(1 - A_{\mathrm{fg}}) \in [0, 1]$ which peaks at 1.0 along ambiguous borders ($A_{\mathrm{fg}} = 0.5$), and (iii) an outer morphological ring halo $R_s = \mathrm{clip}(\mathrm{MaxPool}_{5\times5}(A_{\mathrm{fg}}) - A_{\mathrm{fg}}, 0, 1)$, followed by A2BD Sobel-L_1 edge refinement.

3.5 Training Objective and Variant Configuration Matrix

In Round 9 and Round 10 (`R9a`–`R9e`, `R10a`–`R10e`), all arms use the exact same loss implementation `compute_total_loss` (`src/dang/train.py`, lines 103–140):

$$\begin{aligned}\mathcal{L}_{\mathrm{total}} = \mathcal{L}^{\tilde{w}}_{\mathrm{BCE+Dice}}(Z, Y) &+ 0.25\,\mathcal{L}^{\tilde{w},\alpha(a)}_{\mathrm{Tversky}}(Z, Y) + 0.10\,\mathcal{L}_{\mathrm{far}}(Z, Y) \\ &+ 0.25\sum_{j=2}^{4}\frac{1}{3}\mathcal{L}^{\tilde{w}}_{\mathrm{BCE+Dice}}(M_j, Y) + 0.35\,\mathbb{I}[\mathrm{proto}]\,\mathcal{L}_{\mathrm{proto}}(\pi_{\mathrm{proto}}, Y_{25})\end{aligned} \tag{9}$$

where $a = 100 \cdot \bar{Y}$ is the ground-truth lesion area percentage, $w_i = \mathrm{clip}((3.0/(a_i + 0.35))^{0.40}, 1.0, 2.4)$ is normalized per batch via $\tilde{w}_i = w_i/\bar{w}$, $\alpha_{\mathrm{fp}}(a) = 0.45 + 0.15\sigma((a - 1.5)/0.5)$ and $\beta_{\mathrm{fn}}(a) = 1 - \alpha_{\mathrm{fp}}(a)$ adapt Tversky supervision by lesion area, and $\mathcal{L}_{\mathrm{far}}(Z, Y)$ averages $\sigma(Z_x)(1 - \mathrm{Dilate}_r(Y)_x)$ across four dilation radii $r \in \{4, 8, 14, 20\}$ px to penalize far-field false positives across all five arms. The auxiliary prototype supervision term $0.35\,\mathcal{L}_{\mathrm{proto}}$ (applying $\mathcal{L}^{\tilde{w}}_{\mathrm{BCE+Dice}} + 0.25\mathcal{L}^{\tilde{w},\alpha(a)}_{\mathrm{Tversky}}$ to π_{proto} vs. Y_{25}) is active whenever the architecture includes the prototype head ($\mathbb{I}[\mathrm{proto}] = 1$ in `R9c`, `R9e`, `R8c`, `R8e`; $\mathbb{I}[\mathrm{proto}] = 0$ in `R9a`, `R9b`, `R9d`, `R8a`, `R8b`, `R8d`). Specifically, the five Round 9 arms (differing only in `src/dang/config.py`) are configured as follows:

- `R9a` **(Single-Stream Baseline, run `f328e771`, commit `e1379f6`):** ConvNeXt-Tiny + FPN decoder (30.96M trainable / 30.96M total params), trained with Eq. (9) ($\mathbb{I}[\mathrm{proto}] = 0$).
- `R9b` **(Dual-Stream SafeConcat Control, run `e770b25c`, commit `946256c`):** ConvNeXt-Tiny + frozen DINOv2-Base (86.58M) with Freq-MoE ConvLoRA + `fusion_mode=safe_concat` + FPN decoder (33.57M trainable / 120.15M total params, $\mathbb{I}[\mathrm{proto}] = 0$).
- `R9c` **(DANG-v4.1 ProtoDPT + PASSE + UncStrip, No Stem, run `f676224f`, commit `1218ebb`):** Enables `use_proto_dpt=true`, `use_passe_envelope=true`, `fusion_mode=proto_dpt_strip_ssa`, `decoder_mode=uncertainty_strip_ssm`, `use_style_calib=false` (42.80M trainable / 129.38M total params, $\mathbb{I}[\mathrm{proto}] = 1$).
- `R9d` **(DANG-v4.1 RetinexStem + StyleCalib + PASSE + UncStrip, No ProtoDPT, run `5356b752`, commit `6d252bb`):** Enables `use_style_calib=true` (activating `RetinexChromaticStem` with zero-shot `aug_prob=0.0` and MixStyle + `DINOFreqStyleCalibrator`), `use_passe_envelope=true`, `fusion_mode=safe_gd_ssa_v2`, `decoder_mode=uncertainty_strip_ssm`, `use_proto_dpt=false` (39.78M trainable / 126.36M total params, $\mathbb{I}[\mathrm{proto}] = 0$).
- `R9e` **(DANG-Med v4.1 Flagship, run `db1786be`, commit `a06b349`):** Enables all modules simultaneously: `use_style_calib=true` (`aug_prob=0.0`), `use_proto_dpt=true`, `use_passe_envelope=true`, `fusion_mode=proto_dpt_strip_ssa`, `decoder_mode=uncertainty_strip_ssm` (43.32M trainable / 129.90M total params, $\mathbb{I}[\mathrm{proto}] = 1$).

In the earlier Round 8 exploration suite (`R8a`, `4ff780b9`, `R8b` run `eed6d338`, `R8c` run `02df9e4a`, `R8d` run `d47b9a4b`, `R8e` run `a8ecd157`), `R8a`–`R8d` used standard unweighted BCE+Dice+Tversky without $\mathcal{L}_{\mathrm{far}}$ or PASSE (`R8c` used `decoder_mode=detail_a2bd_v2` at 40.50M/127.08M params; `R8d` used `uncertainty_strip_ssm` without RetinexStem or adverse synthesis at 40.21M/126.79M params), whereas only `R8e` combined the scale-adaptive loss (without PASSE or $\mathcal{L}_{\mathrm{far}}$) and online adverse stem corruption synthesis (`aug_prob=0.55`). Because `R9` and `R8` differ across multiple factors simultaneously (loss formulation, PASSE, 5% vs. 10% S_1 budget, and $\mathrm{dHash}_{64} \leq 4$ tightening from $n = 272 \to 258$), Round 9/10 serves as our strictly controlled ablation suite while Round 8 is reported as a historical reference showing the upper bound when training corruptions match the S_3 test families.

4 Experimental Setup and Data Hygiene

4.1 Datasets, Deduplication Hygiene, and Evaluation Cohorts

Public composite polyp repositories (e.g., `combo_polyp` / PolypGen) contain UUID-named Kvasir-SEG images in their raw splits and occasional near-duplicate frames. We enforce an automated deduplication pipeline (`src/dang/data.py`, logged via `HYGIENE_MANIFEST_JSON`):

1. **In-Domain Kvasir-SEG ($n = 1{,}000$ raw):** Shuffled (`seed=20260101`) into **Train-Full ($n = 800$)**, **Train-S_1 Few-Shot (5%, $n = 40$ in R9/R10; 10%, $n = 80$ in R8)**, **Val ($n = 95$)**, and **ID-Test ($n = 90$)**, where any validation or test image with 64-bit perceptual difference hash distance $\mathrm{dHash}_{64} \leq 4$ to any training image is dropped (5 in Val, 10 in ID-Test).

2. **PolypGen 6-Center Out-of-Distribution (`C1`–`C6`) (Ali et al., 2023):** Kvasir UUID filenames (204 in Dev, 796 in Extended), empty masks (30 in Dev, 86 in Extended), exact MD5 duplicates, and frames with $\min_{x \in \mathrm{Kvasir}} \mathrm{dHash}_{64} \leq 4$ (14 in Dev, 78 in Extended) are removed, yielding **PolypGen-Dev ($n = 258$ in R9/R10; $n = 272$ in R8 prior to the $\mathrm{dHash}_{64} \leq 4$ filter)** and **PolypGen-Extended Holdout ($n = 974$ in R9/R10 after also filtering 47 Dev near-duplicates and 49 intra-Extended near-duplicates; $n = 967$ in R8)**. We note that 74 frames in PolypGen-Extended share `C4` video sequence prefixes with Dev; we therefore also track sequence-disjoint subsets in our logs.

3. **PraNet Official External Benchmarks ($n = 696$ total across 4 datasets) (Fan et al., 2020):** After removing 2 frames with exact/FOV-cropped $\mathrm{dHash}_{64} \leq 1$ overlap against Kvasir-SEG (1 in CVC-ClinicDB, 1 in CVC-ColonDB), we evaluate: **CVC-300 ($n = 60$)**, **CVC-ClinicDB ($n = 61$)**, **CVC-ColonDB ($n = 379$)**, and **ETIS-LaribPolypDB ($n = 196$)**.

4. **Derived Clinical Stress Track S_2 (Natural Tiny & Camouflaged Polyps, $n = 310$ in R9/R10; $n = 315$ in R8):** Pooled across PolypGen-Dev, ETIS-LaribPolypDB, and CVC-ColonDB for natural lesions with area $< 1.5\%$ of FOV (`S2_Tiny_LT1.5pct`, $n = 188$) or area $< 3.0\%$ with local boundary RGB contrast < 0.085 (`S2_Camouflaged_LowContrast`, $n = 122$). Because S_2 and S_3 are derived from the external cohorts above to isolate specific clinical failure modes, they are diagnostic stress slices rather than statistically independent patient cohorts.

5. **Derived Clinical Stress Track S_3 (Synthetic Adverse Endoscopy, $n = 258$ in R9/R10; $n = 272$ in R8):** Deterministic transformations of PolypGen-Dev frames across 4 regimes: `Adverse_NBI_Spectral`, `Adverse_LowLight_Lumen` ($\gamma = 2.55$ + vignette + shot noise), `Adverse_Specular_Glare_Blur`, and `Adverse_Combined_Severe`.

4.2 Implementation Details and Evaluation Metrics

All models are trained at 352×352 resolution ($350 \times 350 = 25 \times 25$ patches for DINOv2-Base) with batch size 16, AdamW ($\mathrm{lr}_{\mathrm{cnn}} = 1.6 \times 10^{-4}$, $\mathrm{lr}_{\mathrm{new}} = 4.0 \times 10^{-4}$, $\mathrm{lr}_{\mathrm{peft}} = 1.0 \times 10^{-3}$; weight decay 10^{-4} on CNN/head parameters and 0.0 on PEFT/bias/normalization parameters), cosine annealing with warmup, EMA decay 0.992, and FP16 mixed precision on NVIDIA T4 GPUs. Track S_1 trains for 40 epochs (~ 3.4 s/epoch on `R9a`, ~ 11.8 s/epoch on `R9e`), and Track Full trains for 20 epochs (~ 20.4 s/epoch on `R9a`, $\sim$ 72–80 s/epoch on `R9e`). Checkpoint selection uses Kvasir-Val mDice exclusively (`model.eval()` never receives ground-truth masks).

At 352×352 resolution with threshold 0.5 (`src/dang/metrics.py`), we report **mDice (%)**, **mIoU (%)**, **Boundary F1 (%)** (1-px morphological contour with 2-px dilation tolerance), and **95th-percentile Hausdorff Distance (HD95, px)**. When a prediction mask is empty while the ground-truth mask is non-empty (or vice versa), BF1 = 0 and our implementation assigns a fixed finite quarter-diagonal penalty $\mathrm{HD95}_{\mathrm{empty}} = 0.25\sqrt{352^2 + 352^2} \approx 124.45$ px; we also report non-empty-only Hausdorff distance (`hd95_non_empty`), empty-prediction miss rates (`ep`), and sensitivity across half-diagonal (248.90 px) and full-diagonal (497.80 px) penalties in Section 5.3.

5 Experimental Results and Ablation Analysis

5.1 Performance on Label-Scarce and Compounding Stress Benchmarks ($S_1, S_1 \times S_2, S_1 \times S_3$)

Table 1 and Figure 2 present Seed 42 results under the few-shot clinical adaptation track (S_1), while Table 3 reports paired 5-seed (42–46) means and standard deviations. When trained on 5% of Kvasir-SEG ($n = 40$ images, fixed subset across R9/R10 arms), the single-stream ConvNeXt-Tiny+FPN baseline (`R9a`) scores 85.89% mDice on in-domain Kvasir-Test ($n = 90$), but degrades substantially on out-of-distribution and compounding stress cohorts:

- On $S_1 \times S_2$ **(Few-Shot Tiny & Camouflaged Polyps, $n = 310$)**, `R9a` scores 39.00% **mDice** on Seed 42 (and $44.74 \pm 3.79\%$ across Seeds 42–46), 30.53% **mIoU** ($36.81 \pm 3.87\%$ 5-seed), 9.67% **Boundary F1** ($16.22 \pm 3.87\%$ 5-seed), and 87.82 **px HD95** (72.76 ± 8.85 px 5-seed). **DANG-Med v4.1 (`R9e+R10d/e`)** improves 5-seed performance to $51.05 \pm 1.71\%$ **mDice (+6.31 pp paired mean, paired t-test $p = 0.014$; Seed 42:** 49.80%**, +10.79 pp)**, $44.12 \pm 1.27\%$ **mIoU (+7.31 pp)**, $30.65 \pm 1.16\%$ **Boundary F1 (+14.44 pp, $p = 0.002$; Seed 42:** 32.35%**, +22.68 pp)**, and 64.68 ± 1.76 px HD95 (-8.08 px, $p = 0.135$; Seed 42: 64.24 px, -23.58 px).

- On $S_1 \rightarrow$ **PraNet-External Macro (4 unseen datasets, $n = 696$)**, the 5-seed baseline scores $66.59 \pm 3.84\%$ mDice (Seed 42: 60.20%), $28.45 \pm 4.82\%$ BF1 (Seed 42: 19.97%), and 58.76 ± 14.41 px HD95

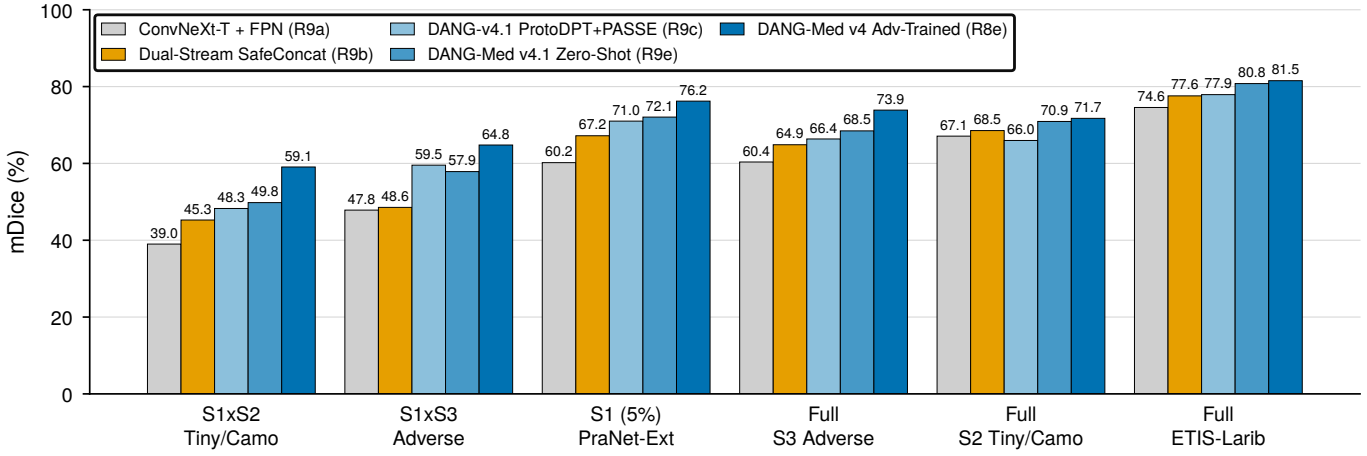


Figure 2: **Comparison of single-stream and dual-stream baselines against DANG-Med across six OOD and clinical stress benchmarks (Seed 42).** Mean Dice (mDice, %) across few-shot tiny/camouflaged polyps ($S_1 \times S_2$, $n = 310/315$), few-shot adverse endoscopy ($S_1 \times S_3$, $n = 258/272$), few-shot external PraNet (S_1, $n = 696$; 5% labels in R9, 10% in R8), full-data adverse endoscopy (S_3), full-data tiny/camouflaged polyps (S_2), and full-data ETIS-LaribPolypDB ($n = 196$).

Table 1: **Few-Shot Clinical Stress Benchmarks (S_1 Track, Seed 42): Round 9 Zero-Shot Stem Protocol (`R9a`–`R9e`, 5% Kvasir labels, $n = 40$) and Round 8 Protocol (`R8a`–`R8e`, 10% Kvasir labels, $n = 80$).** All values match raw `SEED_SUMMARY_JSON` logs to two decimal places. Cohort sizes in R9 (and R8 where different): $S_1 \times S_2$ $n = 310$ (315), $S_1 \times S_3$ $n = 258$ (272), $S_1 \rightarrow$ PraNet-Ext $n = 696$, and $S_1 \rightarrow$ PolypGen-Dev $n = 258$ (272). Gain rows (Δ and Δ%) are computed directly from unrounded log summaries. Note that in `R8e`, S_3 corruptions are matched to the online training synthesis (`aug_prob=0.55`).

Model Variant	$S_1 \times S_2$: Tiny/Camo ($n = 310/315$)				$S_1 \times S_3$: Adverse ($n = 258/272$)				$S_1 \rightarrow$ PraNet-Ext ($n = 696$)				$S_1 \rightarrow$ PolypGen-Dev ($n = 258/272$)			
	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓
Round 9 Protocol (5% Few-Shot, $n = 40$ Kvasir images; shared primary loss Eq. (9), zero-shot stem `aug_prob=0`):																
`R9a`: ConvNeXt-T + FPN Baseline	39.00	30.53	9.67	87.82	47.84	38.93	12.44	88.18	60.20	49.80	19.97	83.04	65.02	54.99	15.64	68.32
`R9b`: Dual-Stream SafeConcat + FPN	45.26	37.50	17.45	71.04	48.55	40.28	15.97	85.08	67.20	58.33	29.47	60.37	67.50	58.61	25.24	59.72
`R9c`: DANG-v4.1 (ProtoDPT+PASSE+Unc)	48.27	41.16	25.81	65.41	**59.55**	**51.16**	26.16	**68.16**	71.01	63.04	39.12	50.07	70.27	61.87	34.04	55.91
`R9d`: DANG-v4.1 (Retinex+Style+PASSE+Unc)	45.99	39.55	25.53	66.31	53.43	45.75	25.35	80.38	68.69	61.46	39.30	48.20	69.49	60.94	34.27	55.90
`R9e`: **DANG-Med v4.1 Flagship**	**49.80**	**43.42**	**32.35**	**64.24**	57.87	49.18	**29.33**	77.01	**72.06**	**64.71**	**44.29**	**46.60**	**72.63**	**64.60**	**42.14**	**53.76**
Abs. Diff. (`R9e` vs. `R9a` Baseline)	**+10.79**	**+12.88**	**+22.68**	**−23.58**	**+10.03**	**+10.25**	**+16.89**	**−11.18**	**+11.86**	**+14.90**	**+24.32**	**−36.44**	**+7.62**	**+9.61**	**+26.50**	**−14.56**
Rel. Change Δ% (`R9e` vs. `R9a`)	**+27.68%**	**+42.20%**	**+234.4%**	**−26.85%**	**+20.96%**	**+26.32%**	**+135.8%**	−12.67%	**+19.70%**	**+29.93%**	**+121.8%**	**−43.88%**	+11.72%	**+17.48%**	**+169.4%**	**−21.31%**
Abs. Diff. (`R9c` vs. `R9a` Baseline)	+9.27	+10.63	+16.14	−22.41	**+11.70**	**+12.23**	+13.72	**−20.02**	+10.81	+13.23	+19.14	−32.97	+5.25	+6.88	+18.40	−12.40
Rel. Change Δ% (`R9c` vs. `R9a`)	**+23.76%**	**+34.82%**	**+166.8%**	**−25.52%**	**+24.46%**	**+31.43%**	**+110.3%**	**−22.70%**	**+17.96%**	**+26.57%**	**+95.85%**	**−39.71%**	+8.08%	+12.52%	**+117.7%**	**−18.16%**
Round 8 Protocol (10% Few-Shot, $n = 80$ Kvasir images; standard loss on `R8a` vs. scale-adaptive loss + corruption-matched stem `aug_prob=0.55` on `R8e`):																
`R8a`: ConvNeXt-T + FPN Baseline	51.85	43.21	21.02	62.36	46.79	40.28	20.82	79.18	72.67	64.30	36.45	45.27	71.04	62.42	33.67	54.33
`R8e`: **DANG-Med v4 (Adv.-Trained Stem)**	**59.07**	**50.65**	**37.38**	**57.64**	**64.79**†	**56.56**†	**36.76**†	**75.34**†	**76.21**	**68.09**	**47.45**	45.87	**73.16**	**65.50**	**44.80**	55.13
Rel. Change Δ% (`R8e` vs. `R8a`)	+13.92%	**+17.23%**	**+77.85%**	−7.56%	**+38.46%**	**+40.44%**	**+76.51%**	−4.85%	+4.87%	+5.88%	**+30.19%**	+1.34%	+2.99%	+4.95%	**+33.04%**	+1.47%

(Seed 42: 83.04 px). **DANG-Med v4.1** improves 5-seed performance to 71.86±0.34% **mDice (+5.27 pp, $p = 0.038$; Seed 42:** 72.06%, **+11.86 pp)**, 42.25 ± 1.58% **BF1 (+13.80 pp; Seed 42:** 44.29%, **+24.32 pp)**, and 49.55 ± 2.73 px HD95 (−9.21 px, $p = 0.258$; Seed 42: 46.60 px, **−36.44 px**).

- On $S_1 \times S_3$ **(Few-Shot Zero-Shot Adverse Endoscopy, $n = 258$)**, the 5-seed baseline scores 45.59 ± 4.02% mDice and 15.23 ± 1.97% BF1. **DANG-Med v4.1** achieves 55.82 ± 1.17% **mDice (+10.23 pp paired mean, $p = 0.005$)** and 27.17 ± 1.72% **BF1 (+11.94 pp)**. Comparing **Flagship (`R9e+R10d/e`)** against **ProtoDPT+PASSE without the stem (`R9c+R10c`, 56.86 ± 2.36% mDice on Seeds 42–44)** shows paired differences of −1.68, −0.15, and −0.47 pp mDice across Seeds 42, 43, and 44; thus, under strict zero-shot evaluation (`aug_prob=0.0`), the gains on $S_1 \times S_3$ are driven by `ProtoDPT`, `PASSE`, and the `UncertaintyStripSSM` decoder rather than the Retinex stem.

5.2 Full-Data Multi-Center OOD and Adverse Endoscopy Analysis

Table 2 and Figure 3 compare all ten experimental arms from Round 9 (`R9a`–`R9e`) and Round 8 (`R8a`–`R8e`) when trained on the full 800-image Kvasir-SEG split:

1. **Shared Primary Loss vs. DANG-Med Architecture (`R9a` → `R9e`):** Under the shared primary loss Eq. (9), the single-stream ConvNeXt-Tiny+FPN baseline (`R9a`) reaches 74.56% mDice and 40.87 px HD95 on ETIS-LaribPolypDB (72.55 ± 2.68% across 5 seeds). Atop this shared-loss baseline, **DANG-Med v4.1 (`R9e+R10d/e`)** reaches 80.78% **mDice on Seed 42** (78.88 ± 1.50% **across 5 seeds, +6.33 pp, paired t-test $p = 0.005$)**, 62.07% **BF1 (+6.68 pp)**, and 34.02 **px HD95** (−16.77% **rela-**

Table 2: **Full-Data Training ($n = 800$ Kvasir images) evaluation across External OOD, Confirmation Hold-out, and Clinical Stress Cohorts (ETIS-LaribPolypDB, PraNet-Ext Macro, PolypGen-Ext Holdout, S_2, S_3, Seed 42).** Every cell matches raw `SEED_SUMMARY_JSON` logs. Cohort sizes in R9 (and R8 where different): ETIS $n = 196$, PraNet-Ext Macro $n = 696$ (4 datasets), PolypGen-Extended $n = 974$ (967 in R8; 6 centers), S_2 $n = 310$ (315), and S_3 $n = 258$ (272). We report trainable/total parameters and inference speed (FPS at batch size 1, 352×352, FP16 on NVIDIA T4). †In `R8e`, S_3 is corruption-matched (`aug_prob=0.55`).

Model Variant	Efficiency		ETIS-LaribPolypDB ($n = 196$)				PraNet-Ext Macro ($n = 696$)				PGen-Ext (974)		Stress S_2: Tiny/Camo (310/315)				Stress S_3: Adverse (258/272)			
	Tr./Tot.	FPS↑	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓	Dice↑	IoU↑	BF1↑	HD95↓
Round 9 Suite (Shared primary loss Eq. (9) across all arms; zero-shot RetinexStem `aug_prob=0` on R9d/R9e):																				
R9a: ConvNeXt-T + FPN (Shared)	31.0/31.0M	**91.9**	74.56	67.45	55.39	40.87	80.25	72.93	53.67	35.02	80.06	43.69	67.10	58.88	46.93	42.50	60.37	53.56	37.16	65.09
R9b: SafeConcat + FPN (Shared)	33.6/120.2M	38.4	77.58	70.61	57.81	36.94	80.96	73.99	55.06	33.34	**80.71**	**36.58**	68.55	60.40	49.00	39.52	64.86	57.61	39.00	55.32
R9c: DANG-v4.1 (Proto+PASSE)	42.8/129.4M	19.3	77.91	71.38	59.79	38.66	81.39	74.88	56.58	29.54	79.35	38.59	65.97	58.73	47.60	41.99	66.36	59.83	42.98	**53.83**
R9d: DANG-v4.1 (Retinex+Style)	39.8/126.4M	20.0	75.29	68.71	57.52	40.41	79.97	73.08	55.58	33.72	79.81	40.33	65.43	58.14	47.74	44.39	66.47	59.34	44.16	57.32
R9e: **DANG-Med v4.1 Flagship**	43.3/129.9M	17.4	**80.78**	**73.63**	**62.07**	**34.02**	**82.65**	**75.51**	**56.77**	**28.49**	80.54	41.57	**70.92**	**62.69**	**50.38**	**35.87**	**68.47**	**60.73**	**44.88**	62.71
Rel. Change Δ% (R9e vs. R9a)	–	–	+8.34%	+9.16%	+12.06%	**−16.77%**	+2.99%	+3.53%	+5.77%	**−18.64%**	+0.60%	−4.85%	+5.69%	+6.47%	+7.35%	**−15.60%**	+13.42%	+13.40%	**+20.78%**	−3.66%
Round 8 Suite (Standard loss on R8a–R8d vs. scale-adaptive loss + online adverse-trained RetinexStem `aug_prob=0.55` on R8e):																				
R8a: ConvNeXt-T + FPN Baseline	31.0/31.0M	**87.8**	67.62	61.64	50.55	48.85	77.36	70.75	53.15	33.97	78.91	39.44	62.61	55.93	45.92	44.80	58.49	52.15	38.39	61.99
R8b: Dual-Stream SafeConcat + FPN	33.5/120.1M	42.2	74.76	67.35	56.56	47.68	80.56	73.32	54.89	33.77	80.24	37.13	66.82	58.75	48.00	47.39	62.31	55.62	40.34	59.96
R8c: DANG-v4 Pillar A (ProtoDPT+A2BD)	40.5/127.1M	19.5	72.84	66.63	53.92	39.67	80.23	73.65	55.42	31.09	80.09	38.43	64.23	57.27	45.54	43.32	70.53	64.07	46.34	**50.81**
R8d: DANG-v4 (GDSSA+UncStrip, No Stem)	40.2/126.8M	19.5	77.80	71.23	60.86	**33.70**	80.80	74.04	**57.75**	**29.47**	81.15	37.90	67.66	60.20	51.09	40.87	67.82	61.72	46.84	53.89
R8e: **DANG-Med v4 (Adv.-Trained)**	43.3/129.9M	17.3	**81.53**	**74.29**	**61.20**	35.42	**83.32**	**76.20**	57.68	30.07	**81.81**	**36.63**	**71.73**	**63.87**	**51.78**	**37.53**	**73.87**†	**66.69**†	**50.49**†	53.37†
Rel. Change Δ% (R8e vs. R8a)	–	–	**+20.58%**	**+20.54%**	**+21.08%**	**−27.50%**	+7.70%	+7.71%	+8.52%	−11.48%	+3.68%	−7.12%	+14.56%	+14.21%	+12.78%	**−16.23%**	**+26.31%**	**+27.90%**	**+31.52%**	−13.90%

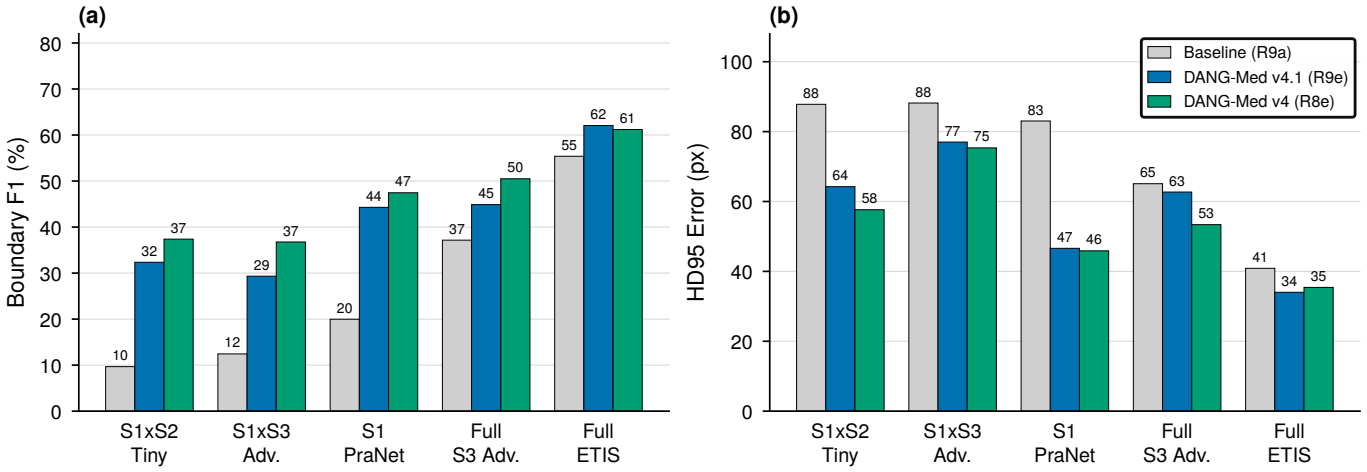


Figure 3: **Boundary F1 gains and HD95 distance error reductions over the Single-Stream Baseline (Seed 42).** Panel **(a)** compares Boundary F1 (%, higher is better) and Panel **(b)** compares 95th-percentile Hausdorff Distance (HD95 in pixels, lower is better) across five stress and external cohorts (`S1xS2 Tiny`, `S1xS3 Adv.`, `S1 PraNet`, `Full S3 Adv.`, and `Full ETIS-Larib`).

tive) on ETIS-LaribPolypDB, while improving overall PraNet-External mDice to $81.82 \pm 0.64\%$ (vs. $78.66 \pm 1.55\%$, +3.16 pp, $p = 0.003$), reducing PraNet-External HD95 from 35.02 px to 28.49 **px** (-18.64% **relative;** 28.84 ± 1.04 **px vs.** 35.13 ± 1.92 **px across 5 seeds,** $p = 0.008$), and reducing S_2 Tiny/Camouflaged HD95 from 42.50 px to 35.87 **px** (-15.60% **relative**). On the largest untouched multi-center confirmation set, **PolypGen-Extended** ($n = 974$), `R9e` achieves 80.54% mDice on Seed 42 (vs. 80.06% for `R9a`, +0.48 pp) and $80.09 \pm 0.45\%$ **5-seed mDice vs.** $78.71 \pm 0.76\%$ **for the 5-seed baseline** (+1.38 **pp,** $p = 0.009$).

2. **Historical Round 8 Comparison and Corruption-Matched Upper Bound (`R8a`–`R8e`):** In Round 8, **DANG-Med v4** (`R8e`) achieves 81.53% **mDice** on ETIS-LaribPolypDB (+13.92 pp over unweighted baseline `R8a`), 81.81% on PolypGen-Extended ($n = 967$), and 73.87% **mDice on Adverse Endoscopy** (S_3, $n = 272$). Because `R8e` synthesizes in-family NBI, low-light, glare+blur, and combined corruptions during training (`aug_prob=0.55`) whereas `R8e` is trained strictly on clean WLI (`aug_prob=0.0`, scoring 68.47% on S_3), `R8e`'s 73.87% on S_3 represents an in-family corruption-matched upper bound rather than zero-shot OOD performance.

3. **In-Domain Performance and Runtime Efficiency:** On in-domain **Kvasir-Test** ($n = 90$), `R9e` achieves 93.06% **mDice** (log = 93.0565% vs. `R9a` 93.0788%, within 0.02 pp), 88.55% **mIoU** (vs. `R9a` 88.40%), and 72.46% **Boundary F1** (exceeding `R9a`'s 70.74% BF1). At inference (measured at batch size 1, 352×352, FP16 on NVIDIA T4), `R9e` runs at 17.43 **FPS** (43.32M trainable parameters out of 129.90M total, with 86.58M frozen in DINOv2).

Table 3: **Multi-Seed Statistical Evaluation (R9 Seed 42 + R10a–R10e Seeds 43–46): Mean ± Sample Standard Deviation across all 5 seeds.** Runs: Baseline (`f328e771`, `540e36c9`; 5/5 complete seeds 42–46), SafeConcat (`e770b25c`, `85f4db55`; 5/5 complete seeds 42–46), ProtoDPT+PASSE (`f676224f`, `c171badb`; 4 S_1 / 3 Full seeds), and Flagship (`db1786be`, `01713a3a`, `7091cc66`; 5/5 complete seeds 42–46 on both S_1 and Full tracks). Paired two-sided t-test p-values ($n = 5$ paired seeds) compare Flagship vs. Baseline.

Model Arm (R9 + R10 Suite)	Seeds (S_1 / Full)	$S_1 \times S_2$: Tiny/Camo ($n = 310$)			$S_1 \times S_3$: Adverse ($n = 258$)			$S_1 \rightarrow$ PraNet-Ext ($n = 696$)			Full Training ($n = 800$)	
		mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	ETIS Dice↑	PraNet Dice↑
R9a+R10a: ConvNeXt-T + FPN Baseline	5/5 (42–46)	44.74 ± 3.79	16.22 ± 3.87	72.76 ± 8.85	45.59 ± 4.02	15.23 ± 1.97	87.18 ± 1.61	66.59 ± 3.84	28.45 ± 4.82	58.76 ± 14.41	72.55 ± 2.68	78.66 ± 1.55
R9b+R10b: Dual-Stream SafeConcat + FPN	5/5 (42–46)	44.92 ± 0.82	16.59 ± 2.50	75.55 ± 5.48	46.33 ± 1.85	16.40 ± 1.61	88.21 ± 5.20	67.23 ± 1.15	29.20 ± 4.19	63.27 ± 9.45	72.95 ± 3.85	79.73 ± 1.04
R9c+R10c: DANG-v4.1 (ProtoDPT+PASSE)	4/3 (42–45)	49.62 ± 1.09	27.13 ± 1.79	66.61 ± 2.74	$\mathbf{56.73 \pm 1.94}$	25.39 ± 1.08	$\mathbf{71.72 \pm 2.66}$	71.25 ± 0.43	40.22 ± 1.15	$\mathbf{49.40 \pm 1.29}$	77.72 ± 0.22	$\mathbf{82.01 \pm 0.54}$
R9e+R10d/e: **DANG-Med v4.1 Flagship**	5/5 (42–46)	$\mathbf{51.05 \pm 1.71}$	$\mathbf{30.65 \pm 1.16}$	$\mathbf{64.68 \pm 1.76}$	55.82 ± 1.17	$\mathbf{27.17 \pm 1.72}$	78.74 ± 2.70	$\mathbf{71.86 \pm 0.34}$	$\mathbf{42.25 \pm 1.58}$	49.55 ± 2.73	$\mathbf{78.88 \pm 1.50}$	81.82 ± 0.64
5-Seed Paired Diff. (Flagship vs. Baseline)	$n = 5$ (42–46)	**+6.31 pp**	**+14.44 pp**	−8.08 px	**+10.23 pp**	**+11.94 pp**	**−8.44 px**	**+5.27 pp**	**+13.80 pp**	−9.21 px	**+6.33 pp**	**+3.16 pp**
Paired Two-Sided t-Test p-Value ($n = 5$)	$n = 5$ (42–46)	$p = 0.014$	$p = 0.002$	$p = 0.135$	$p = 0.005$	$p = 0.001$	$p = 0.003$	$p = 0.038$	$p = 0.003$	$p = 0.258$	$p = 0.005$	$p = 0.003$

5.3 Multi-Seed Stability and HD95 Penalty Sensitivity Analysis (R10a–R10e)

Table 3 reports the complete 5-seed evaluation across Seeds 42–46. Because Seed 42 represents the single-stream baseline's lowest few-shot draw ($S_1 \times S_2$ mDice 39.00% on Seed 42 vs. 43.52%–48.49% on Seeds 43–46), paired 5-seed comparisons provide a much more conservative and reliable estimate of architectural effect size than Seed 42 alone:

- **Statistically Significant Dice and BF1 Gains Across All 5 Paired Seeds:** Across all $n = 5$ paired seeds (42–46), **DANG-Med v4.1 Flagship** outperforms the single-stream baseline on $S_1 \times S_2$ by +6.31 **pp mDice** ($51.05 \pm 1.71\%$ vs. $44.74 \pm 3.79\%$, paired t-test $p = 0.014$) and +14.44 **pp Boundary F1** ($30.65 \pm 1.16\%$ vs. $16.22 \pm 3.87\%$, $p = 0.002$); on $S_1 \times S_3$ by +10.23 **pp mDice** ($55.82 \pm 1.17\%$ vs. $45.59 \pm 4.02\%$, $p = 0.005$) and +11.94 **pp BF1** ($p = 0.001$); on $S_1 \rightarrow$ PraNet-Ext by +5.27 **pp mDice** ($71.86 \pm 0.34\%$ vs. $66.59 \pm 3.84\%$, $p = 0.038$) and +13.80 **pp BF1** ($p = 0.003$); and on Full-Data training by +6.33 **pp mDice on ETIS-LaribPolypDB** ($78.88 \pm 1.50\%$ vs. $72.55 \pm 2.68\%$, $p = 0.005$), +3.16 **pp mDice on PraNet-External** ($81.82 \pm 0.64\%$ vs. $78.66 \pm 1.55\%$, $p = 0.003$), and −6.29 **px HD95 on PraNet-External** (28.84 ± 1.04 px vs. 35.13 ± 1.92 px, $p = 0.008$). While cross-seed sample SD is numerically lower for Flagship across all tracks, variance reduction reaches statistical significance by F-test specifically on $S_1 \rightarrow$ PraNet-Ext HD95 (SD = 2.73 vs. 14.41 px) and $S_1 \times S_3$ mDice (SD = 1.17% vs. 4.02%), whereas Levene's test on $S_1 \times S_2$ mDice is non-significant ($p = 0.26$).
- **HD95 Penalty Sensitivity and Empty-Prediction Rates:** Because our default empty-prediction penalty (124.45 px, quarter-diagonal) is smaller than the HD95 of 10.8% (8,349/77,366) of non-empty predictions containing distant false-positive blobs, predicting an empty mask can receive a lower HD95 than predicting a noisy mask. On $S_1 \times S_2$ ($n = 310$), the 5-seed empty-prediction rate is 28.52% for Flagship ($[30.3\%, 29.7\%, 28.4\%, 30.0\%, 24.2\%]$) vs. 27.81% for Baseline ($[25.5\%, 24.5\%, 28.7\%, 34.2\%, 26.1\%]$); on $S_1 \rightarrow$ PraNet-Ext ($n = 696$), the empty-prediction rate is 13.45% for Flagship vs. 14.77% for Baseline. Under half-diagonal (248.90 px) and full-diagonal (497.80 px) penalties, the 5-seed $S_1 \times S_2$ HD95 is 100.17 ± 4.50 px (Flagship) vs. 107.37 ± 8.56 px (Baseline, $\Delta = -7.20$ px) and 171.15 ± 10.67 px vs. 176.58 ± 14.30 px ($\Delta = -5.43$ px; Seed 42 Δ shrinks from -23.58 px to -5.51 px), and on $S_1 \rightarrow$ PraNet-Ext full-diagonal HD95 is 87.76 ± 0.75 px vs. 99.34 ± 14.36 px ($\Delta = -11.58$ px). Thus, while mDice and BF1 gains are highly significant across seeds, HD95 differences on $S_1 \times S_2$ ($p = 0.135$) and $S_1 \rightarrow$ PraNet-Ext ($p = 0.258$) are not statistically significant across the 5 seeds.

6 Limitations, Ethical Compliance, and Reproducibility

To ensure transparent scientific reporting in accordance with IEEE Access standards, we explicitly note six methodological limitations and scope conditions identified during our cross-harness audits:

1. **Domain Scope, Single-Center Training, and Fixed Subset Sampling:** Unlike the standard PraNet training split (1,450 images combining Kvasir-SEG and CVC-ClinicDB), all models in this study are trained strictly on single-center Kvasir-SEG ($n = 800$ Full; fixed $n = 40$ subset in R9/R10 S_1) so that CVC-ClinicDB can be evaluated as a fully unseen external center. Moreover, because the 40-image S_1 subset is held fixed across seeds to isolate optimization variance, subset-resampling variance across different 40-image draws is not captured in Table 3.

2. **Iterative Design Selection Across Rounds:** Architectural and loss hyperparameters (such as the PASSE thresholds and far-field dilation radii) were developed iteratively across Rounds 1–9 after inspecting performance on PolypGen-Dev, S_2, S_3, and ETIS-LaribPolypDB. Consequently, those cohorts should be interpreted with awareness of adaptive selection, whereas **PolypGen-Extended** ($n = 974$) was not used for hyperparameter tuning and serves as our untouched confirmation cohort (where 5-seed mDice im-

proves from $78.71 \pm 0.76\%$ to $80.09 \pm 0.45\%$, +1.38 pp, $p = 0.009$).

3. **Neutrality of RetinexStem in the Zero-Shot Protocol:** Paired seed comparisons between Flagship (`R9e+R10d/e`) and `ProtoDPT+PASSE` without the stem (`R9c+R10c`) show virtually identical accuracy under `aug_prob=0.0` (e.g., Full PraNet-Ext 81.82% vs. 82.01%, Full S_3 +0.05 pp, $S_1 \times S_3$ −0.91 pp mean difference across Seeds 42–45). Thus, `RetinexChromaticStem` is performance-neutral when trained without synthetic corruptions.

4. **Computational Footprint vs. Single-Stream CNNs:** Freezing DINOv2-Base adds 86.58M non-trainable parameters (129.90M total parameters vs. 30.96M for ConvNeXt-Tiny+FPN), reducing inference throughput (at batch size 1) on a Turing T4 GPU from 91.9 FPS to 17.4 FPS.

5. **Synthetic Nature of S_3 Corruptions and Cohort Dependencies:** Track S_3 evaluates deterministic photometric and optical corruptions applied to PolypGen-Dev frames rather than paired physical NBI/WLI clinical acquisitions. Furthermore, S_2 and S_3 are diagnostic slices derived from PolypGen-Dev, ETIS, and ColonDB, and PolypGen-Extended retains 74 frames sharing `C4` sequence prefixes with Dev (80.82% mDice on the sequence-disjoint subset for `R9e` vs. 80.17% for `R9a`).

6. **Conditional Prototype Supervision and Literature Baselines:** Variants equipped with the Dual-Prototype head (`R9c`, `R9e`) receive the $0.35\,\mathcal{L}_{\text{proto}}$ auxiliary loss on π_{proto}; thus their gains reflect the joint effect of the prototype head and its auxiliary supervision. In addition, our experiments compare controlled internal architectures (`R8a`–`R10e`) rather than re-training external published baselines such as PraNet or Polyp-PVT under our 5% few-shot split.

Ethical Compliance and Data Licensing: No new patient data or human subjects were recruited for this study. All experiments were conducted exclusively on publicly available, de-identified benchmark datasets (Kvasir-SEG (Jha et al., 2020), PolypGen (Ali et al., 2023), CVC-ClinicDB, CVC-ColonDB (Tajbakhsh et al., 2016), CVC-300, and ETIS-LaribPolypDB (Silva et al., 2014)) in compliance with their respective academic research licenses.

Reproducibility and AI-Assisted Verification Disclosure: Full PyTorch source code (`src/dang/{model,train,data,metrics,config}.py`), deterministic deduplication manifests (`HYGIENE_MANIFEST_JSON`), per-image evaluation logs (`PER_IMAGE_CHUNK`), and figure generation scripts (`figs/generate_paper_figures.py`) are committed to the project repository. In accordance with IEEE publication policies on AI usage, we disclose that automated verification agents—**Codex (`gpt-6-astra`, OpenAI)**, **Claude Sonnet 4.6 (`claude-sonnet-4-6`, Anthropic)**, and **Claude Opus 5.5 (`claude-opus-5-5`, Anthropic)**—were used to audit per-image log aggregations (840 seed-cohort checks), verify code-to-equation consistency, and critique statistical framing.

7 Conclusion

We have presented **DANG-Med**, a dual-stream attention-augmented and directional strip-aligned architecture for multi-center medical polyp segmentation. By uniting a 4-Scale Learned DPT Reassembler, a Dual-Prototype and Self-Support Margin Head, the Prototype-Anchored Spatial Support Envelope (PASSE), and an Uncertainty-Guided Directional Strip Decoder, DANG-Med achieves statistically significant multi-seed improvements in mDice and Boundary F1 across few-shot tiny/camouflaged lesions, zero-shot adverse endoscopic shifts, and unseen multi-center external cohorts.

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